

Zcim Project
CP/M® Dynamic Debugging Tool (DDT)
User's Guide

version 2025-08-02

Copyright

Copyright ©1976, 1978 by Digital Research. All rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual or otherwise, without the prior written permission of
~~Digital Research, Post Office Box 579, Pacific Grove, California 93950.~~

~~<http://www.lineo.com>~~

DRDOS, Inc [Bryan Sparks]

Copyright © 2025 by James Burlingame.

License

The original Digital Research Inc assets license is available at

<http://cpm.z80.de/license.html>.

This *Zcim Project edition* is licensed under the
Creative Commons Attribution 4.0 International license. **CC BY 4.0.**

Disclaimer

Digital Research makes no representations or warranties with respect to the contents hereof and specifically disclaims any implied warranties of merchantability or fitness for any particular purpose. Further, Digital Research reserves the right to revise this publication and to make changes from time to time in the content hereof without obligation of Digital Research to notify any person of such revision or changes.

Printing

Zcim Project edition: version 2025-08-02

Contents

1. Introduction	1
2. DDT Commands	3
2.1. The A (Assembly) Command	3
2.2. The D (Display) Command	4
2.3. The F (Fill) Command	5
2.4. The G (Go) Command	5
2.5. The I (Input) Command	6
2.6. The L (List) Command	6
2.7. The M (Move) Command	6
2.8. The R (Read) Command	7
2.9. The S (Set) Command	8
2.10. The T (Trace) Command	8
2.11. The U (Untrace) Command	9
2.12. The X (Examine) Command	9
3. Implementation Notes	10
4. An Example	10
Notes on the <i>Zcim Project</i> Edition	23
License	23

List of Tables

Table 1-1 DDT Line-editing Controls	2
Table 1-2 DDT Commands	2
Table 2-3 CPU Registers	9

1. Introduction

The DDT program allows dynamic interactive testing and debugging of programs generated in the CP/M environment. Invoke the debugger with a command of one of the following forms:

```
DDT
```

```
DDT filename.HEX
```

```
DDT filename.COM
```

where *filename* is the name of the program to be loaded and tested. In both cases, the DDT program is brought into main memory in place of the Console Command Processor (CCP) and resides directly below the Basic Disk Operating System (BDOS) portion of CP/M. Refer to Section 5 for standard memory organization. The BDOS starting address, located in the address field of the JMP instruction at location 0005H, is altered to reflect the reduced Transient Program Area (TPA) size.

The second and third forms of the DDT command perform the same actions as the first, except there is a subsequent automatic load of the specified HEX or COM file. The action is identical to the following sequence of commands:

```
DDT
```

```
Ifilename.HEX or Ifilename.COM
```

```
R
```

where the I and R commands set up and read the specified program to test. See the explanation of the I and R commands below for exact details.

Upon initiation, DDT prints a sign-on message in the form:

```
nnK DDT-S VER m.m
```

where *nn* is the memory size (which much match the CP/M system being used), S is the hardware system which is assumed, corresponding to the codes

D Digital Research standard version

M MDS version

I IMSAI standard version

O Omron systems

S Digital Systems standard version

and *m.m* is the revision number.

Following the sign-on message, DDT prompts the operator with the hyphen character, -, and waits for input commands from the console. The operator can type any of several single character commands, terminated by a carriage return to execute the command. Each line of input can be line-edited using the following standard CP/M controls:

Character	Meaning
rubout/DEL	remove the last character typed
CTRL-U	remove the entire line, ready for re-typing
CTRL-C	system reboot

Table 1-1: DDT Line-editing Controls

Any command can be up to 32 characters in length (an automatic carriage return is inserted as character 33), where the first character determines the command type. Table 1-2 describes DDT commands.

Character	Result
A	enters assembly-language mnemonics with operands.
D	displays memory in hexadecimal and ASCII.
F	fills memory with constant data.
G	begins execution with optional breakpoints.
I	sets up a standard input File Control Block.
L	lists memory using assembler mnemonics.
M	moves a memory segment from source to destination.
R	reads a program for subsequent testing.
S	substitutes memory values.
T	traces program execution.
U	untraced program monitoring.
X	examines and optionally alters the CPU state.

Table 1-2: DDT Commands

The command character, in some cases, is followed by zero, one, two, or three hexadecimal values, which are separated by commas or single blank characters. All DDT numeric output is in hexadecimal form. The commands are not executed until the carriage return is typed at the end of the command.

At any point in the debug run, you can stop execution of DDT by using either a CTRL-C or G0 (jump to location 0000H) and save the current memory image by using a SAVE command of the form:

```
SAVE n filename.COM
```

where *n* is the number of pages (256 byte blocks) to be saved on disk. The number of blocks is determined by taking the high-order byte of the address in the TPA and converting this number to decimal. For example, if the highest address in the TPA is 134H, the number of pages is 12H or 18 in decimal. You could type a CTRL-C during the debug run, returning to the CCP level, followed by

```
SAVE 18 X.COM
```

The memory image is saved as X.COM on the disk and can be directly executed by typing the name *x*. If further testing is required, the memory image can be recalled by typing

```
DDT X.COM
```

which reloads the previously saved program from location 100H through page 18, 23FFH. The CPU state is not a part of the COM file; thus, the program must be restarted from the beginning to test it properly.

2. DDT Commands

The individual commands are detailed below. In each case, the operator must wait for the hyphen prompt character before entering the command. If control is passed to a program under test, and the program has not reached a breakpoint, control can be returned to DDT by executing a RST 7 from the front panel. In the explanation of each command, the command letter is shown in some cases with numbers separated by commas, the numbers are represented by lower-case letters. These numbers are always assumed to be in a hexadecimal radix and from one to four digits in length. Longer numbers are automatically truncated on the right.

Many of the commands operate upon a “CPU state” that corresponds to the program under test. The CPU state holds the registers of the program being debugged and initially contains zeros for all registers and flags except for the program counter, *P*, and stack pointer, *S*, which default to 100H. The program counter is subsequently set to the starting address given in the last record of a HEX file if a file of this form is loaded, see the *I* and *R* commands.

2.1. The A (Assembly) Command

DDT allows in-line assembly language to be inserted into the current memory image using the *A* command, which takes the form:

```
AS
```

where *s* is the hexadecimal starting address for the in-line assembly. DDT prompts the console with the address of the next instruction to fill and reads the console, looking for assembly-language mnemonics followed by register references and operands in absolute hexadecimal form. See the Intel 8080 Assembly Language Reference Card for a list of mnemonics. Each successive load address is printed before reading the console. The A command terminates when the first empty line is input from the console.

Upon completion of assembly language input, you can review the memory segment using the DDT disassembler (see the L command).

Note that the assembler/disassembler portion of DDT can be overlaid by the transient program being tested, in which case the DDT program responds with an error condition when the A and L commands are used.

2.2. The D (Display) Command

The D command allows you to view the contents of memory in hexadecimal and ASCII formats. The D command takes the forms:

D
DS
DS, *f*

In the first form, memory is displayed from the current display address, initially 100H, and continues for 16 display lines. Each display line takes the following form:

```
aaaa bb bb bb bb bb bb bb bb bb bb bb bb bb bb bb ccccccccccccccc
```

where *aaaa* is the display address in hexadecimal and *bb* represents data present in memory starting at *aaaa*. The ASCII characters starting at *aaaa* are to the right (represented by the sequence of character *c*) where non-graphic characters are printed as a period. You should note that both upper- and lower-case alphabetic characters are displayed, and will appear as upper-case symbols on a console device that supports only upper-case. Each display line gives the values of 16 bytes of data, with the first line truncated so that the next line begins at an address that is a multiple of 16.

The second form of the D command is similar to the first, except that the display address is first set to address *s*.

The third form causes the display to continue from address *s* through address *f*. In all cases, the display address is set to the first address not displayed in this command, so that a continuing display can be accomplished by issuing successive D commands with no explicit addresses.

Excessively long displays can be aborted by pressing the return key.

2.3. The F (Fill) Command

The F command takes the form:

FS, f, c

where s is the starting address, f is the final address, and c is a hexadecimal byte constant. DDT stores the constant c at address s , increments the value of s and test against f . If s exceeds f , the operation terminates, otherwise the operation is repeated. Thus, the fill command can be used to set a memory block to a specific constant value.

2.4. The G (Go) Command

A program is executed using the G command, with up to two optional breakpoint addresses. The G command takes the forms:

G

GS

GS, b

GS, b, c

G, b

G, b, c

The first form executes the program at the current value of the program counter in the current machine state, with no breakpoints set. The only way to regain control in DDT is through a RST 7 execution. The current program counter can be viewed by typing an X or XP command.

The second form is similar to the first, except that the program counter in the current machine state is set to address s before execution begins.

The third form is the same as the second, except that program execution stops when address b is encountered (b must be in the area of the program under test). The instruction at location b is not executed when the breakpoint is encountered.

The fourth form is identical to the third, except that two breakpoints are specified, one at b and the other at c . Encountering either breakpoint causes execution to stop and both breakpoints are cleared. The last two forms take the program counter from the current machine state and set one and two breakpoints, respectively.

Execution continues from the starting address in real-time to the next breakpoint. There is no intervention between the starting address and the break address by DDT. If the program under test does not reach a breakpoint, control cannot return to DDT without executing a RST 7 instruction. Upon encountering a breakpoint, DDT stops execution and types

$*d$

where *d* is the stop address. The machine state can be examined at this point using the x (Examine) command. You must specify breakpoints that differ from the program counter address at the beginning of the G command. Thus, if the current program counter is 1234H, then the following commands:

G,1234

and

G400,400

both produce an immediate breakpoint without executing any instructions.

2.5. The I (Input) Command

The I command allows you to insert a filename into the default File Control Block (FCB) at 5CH. The FCB created by CP/M for transient programs is placed at this location (see the CP/M Interface Guide). The default FCB can be used by the program under test as if it had been passed by the CP/M Console Processor. Note that this filename is also used by DDT for reading additional HEX and COM files. The I command takes the forms:

I *filename*

or

I *filename . filetype*

If the second form is used and the filetype is either HEX or COM, subsequent R commands can be used to read the pure binary or hex format machine code. Section 2.8 gives further details.

2.6. The L (List) Command

The L command is used to list assembly-language mnemonics in a particular program region. The L command takes the forms:

L

LS

LS, *f*

The first form lists twelve lines of disassembled machine code from the current list address. The second form sets the list address to *s* and then lists twelve lines of code. The last form lists disassembled code from *s* through address *f*. In all three cases, the list address is set to the next unlisted location in preparation for a subsequent L command. Upon encountering an execution breakpoint, the list address is set to the current value of the program counter (G and T commands). Again, long type-outs can be aborted by pressing RETURN during the list process.

2.7. The M (Move) Command

The M command allows block movement of program or data areas from one location to another in memory. The M command takes the form:

MS, f, d

where s is the start address of the move, f is the final address, and d is the destination address. Data is first removed from s to d , and both addresses are incremented. If s exceeds f , the move operation stops; otherwise, the move operation is repeated.

2.8. The R (Read) Command

The R command is used in conjunction with the I command to read COM and HEX files from the disk into the transient program area in preparation for the debug run. The R command takes the forms:

R

R*b*

where b is an optional bias address that is added to each program or data address as it is loaded. The load operation must not overwrite any of the system parameters from 000H through 0FFH (that is, the first page of memory). If b is omitted, then $b = 0000$ is assumed. The R command requires a previous I command, specifying the name of a HEX or COM file. The load address for each record is obtained from each individual HEX record, while an assumed load address of 100H is used for COM files. Note that any number of R commands can be issued following the I command to reread the program under test, assuming the tested program does not destroy the default area at 5CH. Any file specified with the filetype COM is assumed to contain machine code in pure binary form (created with the LOAD or SAVE command), and all others are assumed to contain machine code in Intel hex format (produced, for example, with the ASM command).

Recall that the command,

DDT *filename.typ*

which initiates the DDT program, equals to the following commands:

DDT

- *filename.typ*

- R

Whenever the R command is issued, DDT responds with either the error indicator ? (file cannot be opened, or a checksum error occurred in a HEX file) or with a load message. The load message takes the form:

NEXT PC

nnnn pppp

where *nnnn* is the next address following the loaded program and *pppp* is the assumed program counter (100H for COM files, or taken from the last record if a HEX file is specified).

2.9. The S (Set) Command

The s command allows memory locations to be examined and optionally altered. The s command takes the form:

ss

where s is the hexadecimal starting address for examination and alteration of memory. DDT responds with a numeric prompt, giving the memory location, along with the data currently held in memory. If you type a carriage return, the data is not altered. If a byte value is typed, the value is stored at the prompted address. In either case, DDT continues to prompt with successive addresses and values until you type either a period or an invalid input value is detected.

2.10. The T (Trace) Command

The T command allows selective tracing of program execution for 1 to 65535 program steps. The T command takes the forms:

T

Tn

In the first form, the CPU state is displayed and the next program step is executed. The program terminates immediately, with the termination address displayed as

*hhhh

where hhhh is the next address to execute. The display address (used in the D command) is set to the value of H and L, and the list address (used in the L command) is set to hhhh. The CPU state at program termination can then be examined using the x command.

The second form of the T command is similar to the first, except that execution is traced for n steps (n is a hexadecimal value) before a program breakpoint occurs. A breakpoint can be forced in the trace mode by typing a rubout character. The CPU state is displayed before each program step is taken in trace mode. The format of the display is the same as described in the x command.

You should note that program tracing is discontinued at the CP/M interface and resumes after return from CP/M to the program under test. Thus, CP/M functions that access I/O devices, such as the disk drive, run in real-time, avoiding I/O timing problems. Programs running in trace mode execute approximately 500 times slower than real-time because DDT gets control after each user instruction is executed. Interrupt processing routines can be traced, but commands that use the breakpoint facility (G, T, and U) accomplish the break using an RST 7 instruction, which means that the tested program cannot use this interrupt location. Further, the trace mode always runs the tested program with interrupts enabled, which may cause problems if asynchronous interrupts are received during tracing.

To get control back to DDT during trace, press RETURN rather than executing an RST 7. This ensures that the trace for current instruction is completed before interruption.

2.11. The U (Untrace) Command

The u command is identical to the T command, except that intermediate program steps are not displayed. The untrace mode allows from 1 to 65535 (0FFFFH) steps to be executed in monitored mode and is used principally to retain control of an executing program while it reaches steady state conditions. All conditions of the T command apply to the u command.

2.12. The X (Examine) Command

The x command allows selective display and alteration of the current CPU state for the program under test. The x command takes the forms:

x

xr

where *r* is one of the 8080 CPU registers listed in Table 2-3.

Register	Meaning	Value
C	Carry flag	0 or 1
Z	Zero flag	0 or 1
M	Minus flag	0 or 1
E	Even parity flag	0 or 1
I	Half-carry	0 or 1
A	Accumulator	0 ... FF
B	BC register pair	0 ... FFFF
D	DE register pair	0 ... FFFF
H	HL register pair	0 ... FFFF
S	Stack pointer	0 ... FFFF
P	Program counter	0 ... FFFF

Table 2-3: CPU Registers

In the first case, the CPU register state is displayed in the format:

```
cf zf mf ef if A=bb B=dddd D=dddd H=dddd S=dddd P=dddd inst
```

where *f* is a 0 or 1 flag value, *bb* is a byte value, and *dddd* is a double-byte quantity corresponding to the register pair. The *inst* field contains the disassembled instruction, that occurs at the location addressed by the CPU state's program counter.

The second form allows display and optional alteration of register values, where *r* is one of the registers given above (C, Z, M, E, I, A, B, D, H, S, or P). In each case, the flag or register value is first displayed at the console. The DDT program then accepts input from the console. If a carriage return is typed, the flag or register value is not altered. If a value in the proper range is typed, the flag or register value is altered. You should note that BC, DE, and HL are displayed as register pairs. Thus, you must type the entire register pair when B, C, or the BC pair is altered.

3. Implementation Notes

The organization of DDT allows certain nonessential portions to be overlaid to gain a larger transient program area for debugging large programs. The DDT program consists of two parts: the DDT nucleus and the assembler/disassembler module. The DDT nucleus is loaded over the CCP and, although loaded with the DDT nucleus, the assembler/disassembler can be overlaid unless used to assemble or disassemble.

In particular, the BDOS address at location 0006H (address field of the JMP instruction at location 0005H) is modified by DDT to address the base location of the DDT nucleus, which, in turn, contains a JMP instruction to the BDOS. Thus, programs that use this address field to size memory see the logical end of memory at the base of the DDT nucleus rather than the base of the BDOS.

The assembler/disassembler module resides directly below the DDT nucleus in the transient program area. If the A, L, T, or X commands are used during the debugging process, the DDT program again alters the address field at 0006H to include this module, further reducing the logical end of memory. If a program loads beyond the beginning of the assembler/disassembler module, the A and L commands are lost (their use produces a ? in response) and the trace and display (T and X) commands list the inst field of the display in hexadecimal, rather than as a decoded instruction.

4. An Example

The following example shows an edit, assemble, and debug for a simple program that reads a set of data values and determines the largest value in the set. The largest value is taken from the vector and stored into LARGE at the termination of the program.


```

: *BOP↵
1:          ORG          100H          ;START OF TRANSIENT
2:                                     ;AREA
3:          MVI          B, LEN        ;LENGTH OF VECTOR TO SCAN
4:          MVI          C, 0          ;LARGER_RST VALUE SO FAR
5:          LXI          H, VECT       ;BASE OF VECTOR
6:  LOOP:    MOV          A, M          ;GET VALUE
7:          SUB          C             ;LARGER VALUE IN C?
8:          JNC          NFOUND        ;JUMP IF LARGER VALUE NOT
9:                                     ;FOUND
10: ;
11:          NEW LARGEST VALUE, STORE IT TO C
12:          MOV          C, A
13:  NFOUND   INX          H             ;TO NEXT ELEMENT
14:          DCR          B             ;MORE TO SCAN?
15:          JNZ          LOOP         ;FOR ANOTHER
16: ;
17:          END OF SCAN, STORE C
18:          MOV          A, C          ;GET LARGEST VALUE
19:          STA          LARGE
20:          JMP          0             ;REBOOT
21: ;
22:          TEST DATA
23:  VECT:    DB          2,0,4,3,5,6,1,5
24:  LEN      EQU          $-VECT       ;LENGTH
1: *E↵ ← End of Edit

```

A>ASM SCAN↵ *Start Assembler*
CP/M ASSEMBLER - VER 1.0

0122
002H USE FACTOR
END OF ASSEMBLY *Assembly Complete - Look at Program Listing*

A>TYPE SCAN.PRN↓

Look at Program Listing

Code Address		Machine Code	Source Program		
0100			ORG	100H	;START OF TRANSIENT
					;AREA
0100	0608		MVI	B, LEN	;LENGTH OF VECTOR TO SCAN
0102	0E00		MVI	C, 0	;LARGER_RST VALUE SO FAR
0104	211901		LXI	H, VECT	;BASE OF VECTOR
0107	7E	LOOP:	MOV	A, M	;GET VALUE
0108	91		SUB	C	;LARGER VALUE IN C?
0109	D20D01		JNC	NFOUND	;JUMP IF LARGER VALUE NOT
					;FOUND
					NEW LARGEST VALUE, STORE IT TO C
010C	4F		MOV	C, A	
010D	23	NFOUND	INX	H	;TO NEXT ELEMENT
010E	05		DCR	B	;MORE TO SCAN?
010F	C20701		JNZ	LOOP	;FOR ANOTHER
					END OF SCAN, STORE C
0112	79		MOV	A, C	;GET LARGEST VALUE
0113	322101		STA	LARGE	
0116	C30000		JMP	0	;REBOOT
					TEST DATA
0119	0200040305	VECT:	DB	2,0,4,3,5,6,1,5	
0008	=	LEN	EQU	\$-VECT	;LENGTH
0121		LARGE:	DS	1	;LARGEST VALUE ON EXIT
0122			END		

Code/Data listing ;

truncated ↓ ;

value of Equate

A>DDT SCAN.HEX↵

Start Debugger with hex format machine code

16K DDT VERS 1.0

NEXT PC

0121 0000

-X↵ last load address + 1

next instruction to execute
at PC = 0

COZOM0E0IO A=00 B=0000 D=0000 H=0000 S=0100 P=0000 JMP FA03

examine registers before debug run

-XP↵

P=0000 100↵ change PC to 100

-X↵ look at registers again

PC changed

COZOM0E0IO A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08

next instruction
to execute at PC=100

-L100↵

0100 MVI B,08
0102 MVI C,00
0104 LXI H,0119
0107 MOV A,M
0108 SUB C
0109 JNC 010D
010C MOV C,A
010D INX H
010E DCR B
010F JNZ 0107
0112 MOV A,C

Disassembled machine code
at 100H
(See source listing
for comparison)

-L↵

0113 STA 0121
0116 JMP 0000
0119 STAX B
011A NOP
011B INR B
011C INX B
011D DCR B
011E MVI B,01
0120 DCR B
0121 ??= 20
0122 MOV D,D

A little more
machine code
(note that program
ends at location 116
with a JMP to 0000)

-A116 ↵

0116 RST 7 ↵

0117 ↵

-L113 ↵

enter inline assembly mode to change the JMP to 0000 into a RST 7, which will cause the program under test to return to DDT if 116H is ever executed (single carriage return stops assemble mode)

list code at 113H to check that RST 7 was properly inserted in place of JMP

```
0113 STA 0121
0116 RST 07
0117 NOP
0118 NOP
0119 STAX B
011A NOP
011B INR B
011C INX B
011D DCR B
011E MVI B,01
0120 DCR B
```

-X ↵ look at registers

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08

-T ↵ execute program for one step

initial CPU state, before instruction is executed

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08*0102

-T ↵ trace one step again (note 08H in B)

automatic breakpoint

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0102 MVI C,00*0104

-T ↵ trace again (register C is cleared)

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0104 LXI H,0119*0107

-T3 ↵ trace three steps

C0Z0M0E0I0 A=00 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C

C0Z0M0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JNC 010D*010D

automatic breakpoint at 10DH

	program data								lowercase x															
0119	02	00	04	03	05	06	01																	
0120	05	11	00	22	21	00	02	7E	EB	77	13	23	EB	0B	78	B1	...	!"!...~.W.#...	X					
0130	C2	27	01	C3	03	29	00	00	00	00	00	00	00	00	00	00	00	00	00					
0140	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
0150	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
0160	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
0170	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
0180	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
0190	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
01A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
01B0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					
01C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00					

```
COZOMOEIO A=02 B=0800 D=0000 H=0119 S=0100 P=010D INX H
```

C0Z0M0E0I0	A=02	B=0800	D=0000	H=0119	S=0100	P=010D	INX	H
C0Z0M0E0I0	A=02	B=0800	D=0000	H=011A	S=0100	P=010E	DCR	B
C0Z0M0E0I0	A=02	B=0700	D=0000	H=011A	S=0100	P=010F	JNZ	0107
C0Z0M0E0I0	A=02	B=0700	D=0000	H=011A	S=0100	P=0107	MOV	A,M
C0Z0M0E0I0	A=00	B=0700	D=0000	H=011A	S=0100	P=0108	SUB	C*0109

C0Z1M0E0I0 A=00 B=0700 D=0000 H=011A S=0100 P=0109 JNC 010D*0108

COZOMOEIO A=04 B=0600 D=0000 H=011B S=0100 P=0108 SUB C

*0116 *breakpoint at 116H, caused by executing RST 7 in machine code*

C0Z1M0E0I0 A=00 B=0000 D=0000 H=0121 S=0100 P=0116 RST 07

P=0116 100↓

```
COZ1M0E0I0 A=00 B=0000 D=0000 H=0121 S=0100 P=0100 MVI B,08
```

-T10↓ trace 10 (hexadecimal) steps

		first data element	current largest value		
C0Z1M0E0I0	A=00	B=0000	D=0000	H=0121	S=0100 P=0100 MVI B,08
C0Z1M0E0I0	A=00	B=0800	D=0000	H=0121	S=0100 P=0102 MVI C,00
C0Z1M0E0I0	A=00	B=0800	D=0000	H=0121	S=0100 P=0104 LXI H,0119
C0Z1M0E0I0	A=00	B=0800	D=0000	H=0119	S=0100 P=0107 MOV A,M
C0Z1M0E0I0	A=02	B=0800	D=0000	H=0119	S=0100 P=0108 SUB C ← subtract for comparison A < C
C0Z0M0E0I0	A=02	B=0800	D=0000	H=0119	S=0100 P=0109 JNC 010D
C0Z0M0E0I0	A=02	B=0800	D=0000	H=0119	S=0100 P=010D INX H
C0Z0M0E0I0	A=02	B=0800	D=0000	H=011A	S=0100 P=010E DCR B
C0Z0M0E0I0	A=02	B=0700	D=0000	H=011A	S=0100 P=010F JNZ 0107
C0Z0M0E0I0	A=02	B=0700	D=0000	H=011A	S=0100 P=0107 MOV A,M
C0Z0M0E0I0	A=00	B=0700	D=0000	H=011A	S=0100 P=0108 SUB C
C0Z1M0E0I0	A=00	B=0700	D=0000	H=011A	S=0100 P=0109 JNC 010D
C0Z1M0E0I0	A=00	B=0700	D=0000	H=011A	S=0100 P=010D INX H
C0Z1M0E0I0	A=00	B=0700	D=0000	H=011B	S=0100 P=010E DCR B
C0Z0M0E0I0	A=00	B=0600	D=0000	H=011B	S=0100 P=010F JNZ 0107
C0Z0M0E0I0	A=00	B=0600	D=0000	H=011B	S=0100 P=0107 MOV A,M*0108

-A109↓ Insert a "hot patch" into the machine

-JC 10D↓ code to change the JNC to JC

010C↓ stop DDT so that a version of the

-G0↓ patched program can be saved

A>SAVE 1 SCAN.COM↓ program resides on first page, so save 1 page

A>DDT SCAN.COM↓ restart DDT with the saved memory image to continue testing

DDT VERS 2.2

NEXT PC

0200 0100

Program should have moved the value from A into C since A > C. Since this case was not executed, it appears that the JNC should have been a JC instruction

-L100↓ list some code

0100	MVI	B,08
0102	MVI	C,00
0104	LXI	H,0119
0107	MOV	A,M
0108	SUB	C
0109	JC	010D ← previous patch is present in SCAN.COM
010C	MOV	C,A
010D	INX	H
010E	DCR	B
010F	JNZ	0107
0112	MOV	A,C

-XP↓

P=0100 ↓

-T10 ↴ trace to see how patched version operates

data is moved from A to C

```

C0Z0M0E0I0 A=00 B=0000 D=0000 H=0000 S=0100 P=0100 MVI B,08
C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0102 MVI C,00
C0Z0M0E0I0 A=00 B=0800 D=0000 H=0000 S=0100 P=0104 LXI H,0119
C0Z0M0E0I0 A=00 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M
C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C
C0Z0M0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JC 010D
C0Z0M0E0I1 A=02 B=0800 D=0000 H=0119 S=0100 P=010C MOV C,A
C0Z0M0E0I1 A=02 B=0802 D=0000 H=0119 S=0100 P=010D INX H
C0Z0M0E0I1 A=02 B=0802 D=0000 H=011A S=0100 P=010E DCR B
C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=010F JNZ 0107
C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=0107 MOV A,M
C0Z0M0E0I1 A=00 B=0702 D=0000 H=011A S=0100 P=0108 SUB C
C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=0109 JC 010D
C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=010D INX H
C1Z0M1E0I0 A=FE B=0702 D=0000 H=011B S=0100 P=010E DCR B
C1Z0M0E1I1 A=FE B=0602 D=0000 H=011B S=0100 P=010F JNZ 0107*0107

```

breakpoint after 16 steps ↗

-X ↴

C1Z0M0E1I1 A=FE B=0602 D=0000 H=011B S=0100 P=0107 MOV A,M

-G,108 ↴ run from current PC and breakpoint at 108H

*0108

-X ↴

next data item
C1Z0M0E1I1 A=04 B=0602 D=0000 H=011B S=0100 P=0108 SUB C

-T ↴

single step for a few cycles

C1Z0M0E1I1 A=04 B=0602 D=0000 H=011B S=0100 P=0108 SUB C*0109

-T ↴

C0Z0M0E0I1 A=02 B=0602 D=0000 H=011B S=0100 P=0109 JC 010D*010C

-X ↴

C0Z0M0E0I1 A=02 B=0602 D=0000 H=011B S=0100 P=010C MOV C,A

-G ↴

run to completion

*0116

-X ↴

C0Z1M0E1I1 A=03 B=0003 D=0000 H=0121 S=0100 P=0116 RST 07

-S121 ↴

look at the value of "LARGE"

0121 03 ↴ Wrong value!

0122 52 . ↴

- L100 ↴

```
0100 MVI B,08
0102 MVI C,00
0104 LXI H,0119
0107 MOV A,M
0108 SUB C
0109 JC 010D
010C MOV C,A
010D INX H
010E DCR B
010F JNZ 0107
0112 MOV A,C
```

Review the code

- L ↴

```
0113 STA 0121
0116 RST 07
0117 NOP
0118 NOP
0119 STAX B
011A NOP
011B INR B
011C INX B
011D DCR B
011E MVI B,01
0120 DCR B
```

- XP ↴

reset the PC

P=0116 100 ↴

- T ↴ single step, and watch data values

C0Z1M0E1I1 A=03 B=0003 D=0000 H=0121 S=0100 P=0100 MVI B,08*0102

- T ↴

count set

C0Z1M0E1I1 A=03 B=0803 D=0000 H=0121 S=0100 P=0102 MVI C,00*0104

- T ↴

"largest" set

C0Z1M0E1I1 A=03 B=0800 D=0000 H=0121 S=0100 P=0104 LXI H,0119*0107

- T ↴

base address of data set

C0Z1M0E1I1 A=03 B=0800 D=0000 H=0119 S=0100 P=0107 MOV A,M*0108

- T ↴

first data item brought to A

C0Z1M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0108 SUB C*0109

-T↵

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=0109 JC 010D*010C

-T↵

C0Z0M0E0I0 A=02 B=0800 D=0000 H=0119 S=0100 P=010C MOV C,A*010D

-T↵

first data item moved to C correctly

C0Z0M0E0I1 A=02 B=0802 D=0000 H=0119 S=0100 P=010D INX H*010E

-T↵

C0Z0M0E0I1 A=02 B=0802 D=0000 H=011A S=0100 P=010E DCR B*010F

-T↵

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=010F JNZ 0107*0107

-T↵

C0Z0M0E0I1 A=02 B=0702 D=0000 H=011A S=0100 P=0107 MOV A,M*0108

-T↵

second data item brought to A

C0Z0M0E0I1 A=00 B=0702 D=0000 H=011A S=0100 P=0108 SUB C*0109

-T↵

subtract destroys data value which was loaded!

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=0109 JC 010D*010D

-T↵

C1Z0M1E0I0 A=FE B=0702 D=0000 H=011A S=0100 P=010D INX H*010E

-L100↵

0100	MVI	B,08
0102	MVI	C,00
0104	LXI	H,0119
0107	MOV	A,M
0108	SUB	C
0109	JC	010D
010C	MOV	C,A
010D	INX	H
010E	DCR	B
010F	JNZ	0107
0112	MOV	A,C

This should have been a CMP so that register A would not be changed

-A108↵

0108 **CMP C** *hot patch at 108H changes SUB to CMP*

0109 ↵

-G0↵ *stop DDT for Save*

A>SAVE 1 SCAN.COM↵ *save memory image*

A>DDT SCAN.COM ↵ restart DDT

DDT VERS 2.2

NEXT PC

0200 0100

-XP ↵

P=0100 ↵

-L116 ↵

0116	RST	07
0117	NOP	
0118	NOP	
0119	STAX	B
011A	NOP	

look at code to see if it was properly loaded
(long timeout aborted with rubout)

⌘

-G,116 ↵ run from 100H to completion

*0116

-XC ↵ look at Carry (accidental typo)

C1 ↵

-X ↵ look at CPU state

C1Z1M0E1I1 A=06 B=0006 D=0000 H=0121 S=0100 P=0116 RST 07

-S121 ↵ look at "LARGE" - it appears to be correct

0121 06 ↵

0122 00 ↵

0123 22 . ↵

-G0 ↵

stop DDT

Re-edit the source program, and make both changes

A>ED SCAN.ASM ↵

: *NSUB ↵

7: *0LT ↵

7: SUB C ;LARGER VALUE IN C?

7: *SSUB ↵ ~~ZCMP~~ ↵ ~~Z0LT~~ ↵

7: CMP C ;LARGER VALUE IN C?

7: * ↵

8: JNC NFOUND ;JUMP IF LARGER VALUE NOT FOUND

8: *SNC ↵ ~~ZC~~ ↵ ~~Z0LT~~ ↵

8: JC NFOUND ;JUMP IF LARGER VALUE NOT FOUND

8: *E ↵

A>ASM SCAN.AAZ

Re-assemble. Selection source from disk A

CP/M ASSEMBLER - VER 2.0

hex to disk A

0122

print to Z (selects no file)

002H USE FACTOR

END OF ASSEMBLY

A>DDT SCAN.HEX

Re-run debugger to check changes

DDT VERS 2.2

NEXT PC

0121 0000

-L116

check to ensure end is still at 116H

0116 JMP 0000

0119 STAX B

011A NOP

011B INR B

⌘ (rubout)

-G100,116

Go from beginning with breakpoint at end

*0116

breakpoint reached

-D121

look at "LARGE"

0121 06 00 22 21 00 02 7E EB 77 13 23 EB 0B 78 B1 .. '!... W .#..X.
0130 C2 27 01 C3 03 29 00 00 00 00 00 00 00 00 00 ..'...).....
0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
correct value computed

⌘ (rubout) abort long typeout

-G0

stop DDT, debug session complete

Notes on the Zcim Project Edition

The *Zcim Project* goal is to improve the accessibility of legacy Z80 CP/M era content. As part of the project, Jim Burlingame (jb@samplx.org) created a *Typst* version of this manual.

You can find out more about the project at its site z80cim.org. The sources of this document are available on GitHub.

The source material is from the *Tim Olmstead Memorial Digital Research CP/M Library*

The individual documents from the archive include:

- *CP/M Dynamic Debugging Tool (DDT) User's Guide* (pdf)

The contents of the manual were edited using the Typst.app site.

License

The source documentation is under a license granted by the owner of the Digital Research intellectual property in an email available at <http://cpm.z80.de/license.html>.

The *Zcim Project* edition is under the Creative Commons Attribution 4.0 International license.

CC BY 4.0.